MIL-E-0022200/8(SHIPS)
7 December 1971
USED IN LIEU OF
MIL-E-16589B
29 March 1958

MILITARY SPECIFICATION

ELECTRODES, WELDING, COVERED, LOW-HYDROGEN, AND IRON-POWDER LOW-HYDROGEN, CHROMIUM-MOLYBDENUM ALLOY STEEL AND CORROSION RESISTING STEEL

1. SCOPE

1.1 Scope. This specification covers electrodes for welding alloy steels used for corrosion resistance in certain media, such as petroleum crudes, and for high-temperature strength and resistance to creep (see 6.1).

1.2 Classification. Electrodes shall be furnished in the types, classes, and sizes shown in table I, as specified (see 6.2).

Table I - Types, classes, and sizes[1/].

Type	Class	Positions of welding	Diameter of core wire (Inch)
MIL-8015-B2 MIL-8016-B2 MIL-8018-B2	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4
MIL-8015-B2L MIL-8016-B2L	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4
MIL-9015-B3 MIL-9016-B3 MIL-9018-B3	1 2	All Horizontal fillets and flat	3/32, 1/8 5/32 3/16, 7/32, 1/4
MIL-9015-B3L MIL-9016-B3L	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4
MIL-410-15 MIL-410-16	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4
MIL-502-15 MIL-502-16	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4
MIL-502-15-L MIL-502-16-L	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4
MIL-505-15 MIL-505-16	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4

1/ Type classification of electrodes will be determined according to the capability of the electrode to meet the tests specified herein, when used in the positions of welding specified in table I and in section 4. An electrode shall be approved under ONE type classification; and an electrode shall NOT be approved if it differs in type from that stated in the manufacturer's request for test. The position description indicates the maximum size qualified under this specification. However, reference should be made to the Qualified Products List (QPL) for the selection of electrodes for all positions of welding.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-10430 - Welding Rods and Electrodes; Preparation for Delivery of.
MIL-E-0022200E - Electrodes, Welding, Covered; General Specification for.
MIL-S-22698 - Steel Plate, Carbon, Structural, for Ships.

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STANDARDS

MILITARY

MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-418 - Mechanical Tests for Welded Joints.

PUBLICATION

MILITARY

NAVSHIPS 0900-003-9000 - Radiographic Standards for Production and Repair Welds.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Ill. 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A131 - Structural Steel for Ships.
A370 - Methods and Definitions for Mechanical Testing of Steel Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19101.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

Z49.1-1967 - Safety in Welding and Cutting.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N. Y. 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Electrodes furnished under this specification shall conform to the requirements of MIL-E-0022200E and as specified herein.

3.2 Electrode composition.

3.2.1 Core wire composition. Core wire composition is optional with the manufacturer.

3.2.2 Covering. The chemical composition of the covering, except as specified in 3.2.2.1 and 3.2.2.2 is optional with the manufacturer.

3.2.2.1 Total iron content. The total iron of the covering of MIL-XX18 type electrodes, including any combined iron in addition to metallic iron powder, shall be not less than 15 percent.

3.2.2.2 Water content. The water content of the covering shall not exceed the percent-by-weight values specified in table II.

Table II - Water content.

Type	Percent by weight maximum
MIL-80XX	0.4
MIL-90XX	0.2
MIL-410-XX	0.6
MIL-502-XX	0.6
MIL-505-XX	0.6

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3.3 Deposited weld metal.

3.3.1 Mechanical properties. The mechanical properties of weld metal deposited in groove joints shall be as specified in table III. The stress-relieved properties shall be determined for weld metal, postheat treated in accordance with 4.5.1.1 (see table XI).

Table III - Mechanical properties.

Type	Ultimate tensile strength		Yield strength 1/		Elongation in 2 inches		Charpy-V notch impact stress-relieved	
	As-welded	Stress-relieved	As-welded	Stress-relieved	As-welded	Stress-relieved	Energy ft/lbs	Testing
	P.s.i. (Minimum)	P.s.i. (Minimum)	P.s.i. (Minimum)	P.s.i. (Minimum)	Percent (Minimum)	Percent (Minimum)	Minimum average	Minus °F.
MIL-8015-B2L 2/ MIL-8016-B2L MIL-9015-B3L MIL-9016-B3L MIL-502-15-L MIL-502-16-L	75,000	70,000	50,000	45,000	20 20 18 18 17 17	22	—	—
MIL-8015-B2 3/ MIL-8016-B2	—	80,000	—	67,000	—	19	—	—
MIL-8018-B2 4/	—	80,000	—	67,000	—	19	20	20
MIL-9015-B3 3/ MIL-9016-B3	—	90,000	—	67,000	—	17	—	—
MIL-9018-B3 4/	—	90,000	—	77,000	—	17	20	20
MIL-410-15 3/ MIL-410-16 MIL-502-15 MIL-502-16 MIL-505-15 MIL-505-16	—	75,000	—	50,000	—	20	—	—

1/ The yield strength shall be determined by the offset method (offset = 0.2 percent).

2/ The as-welded and stress-relieved tensile properties shall be determined for qualification testing of electrodes for the QPL. The as-welded tensile properties shall be determined for comparsion inspection, when required, and quality conformance inspection.

3/ The stress-relieved tensile properties shall be determined for qualification testing of electrodes for the QPL. The stress-relieved tensile properties shall be determined for comparison inspection, when required, and quality conformance inspection.

4/ The stress-relieved tensile and impact properties shall be determined for qualification testing of electrodes for the QPL. The stress-relieved tensile properties shall be determined for comparison inspection, when required, and quality conformance inspection.

3.4 Chemical composition of deposited weld metal. Chemical composition of deposited weld metal shall be as specified in table IV.

3.5 Electrode identification. Electrodes shall be identified as specified in 3.5.10 of MIL-E-0022200E.

3.6 Electrode lengths. The specified lengths of end grip electrodes shall be in accordance with table V (see 6.2).

3.7 Decoated core wire. Any core wire or electrode which is removed from the assembly line area during production of a lot of electrodes shall not be used to fill any order under this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions shall be in accordance with MIL-E-0022200E and as specified herein.

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Table IV - Chemical composition of deposited weld metal, percent.[1/]

Type	Carbon	Manganese	Phosphorus	Sulfur	Silicon	Chromium	Molybdenum
MIL-8015-B2 MIL-8016-B2 MIL-8018-B2	0.10	0.90	0.03	0.04	0.60 .60 .80	1.00-1.50	0.40-.65
MIL-8015-B2L MIL-8016-B2L	.05	——	.03	.04	1.00 .90	1.00-1.50	.40-.65
MIL-9015-B3 MIL-9016-B3 MIL-9018-B3	.10	.90	.03	.04	.60 .60 .80	2.00-2.50	.90-1.20
MIL-9015-B3L MIL-9016-B3L	.05	.90	.03	.04	1.00 .90	2.00-2.50	.90-1.20
MIL-410-15 MIL-410-16	.10	1.00	.04	.03	.90	11.00-13.50	——
MIL-502-15 MIL-502-16	.10	1.00	.04	.03	.90	4.00-6.00	.45-.65
MIL-502-15-L MIL-502-16-L	.05	1.00	.04	.03	.90	4.00-6.00	.45-.65
MIL-505-15 MIL-505-16	.10	1.00	.04	.03	.60	8.50-10.50	1.00-1.50

1/ Single values are maximum percentages.

Table V - Electrode lengths.

Size (Inch)	Length (Inches)
3/32	9 or 12
1/8, 5/32	9 or 14
3/16	14
7/32, 1/4	18

4.2 Qualification tests. Qualification tests shall be in accordance with 4.2 of MIL-E-0022200E and tables X and XII herein.

4.3 Comparison inspection. In addition to the examination and tests required in 4.5 of MIL-E-0022200E, tests specified in table VI shall be conducted as specified in 4.3 of MIL-E-0022200E, when required by the Naval Ship Engineering Center (NAVSEC).

Table VI - Comparison inspection.

Electrode type	Sizes	Tests	Requirements
All	All	4.6.8 (MIL-E-0022200E)	3.4 and table IV herein
All	All	Weld test No. 3 (see table X herein)	3.3.1 and table III herein

4.4 Quality conformance inspection.

4.4.1 Lot. A lot of electrodes shall be as specified in 4.4.3 and 4.4.6.2 of MIL-E-0022200E for inspection level B.

4.4.2 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be in accordance with 4.4.7 through 4.4.8.3 of MIL-E-0022200E for tests specified in MIL-E-0022200E where required and as specified in 4.4.3 herein for quality conformance weld tests.

4.4.3 Quality conformance inspection. Quality conformance tests shall be conducted on samples selected as specified in 4.4.2 from electrodes produced from each wet mix covering mixture in each lot as defined in

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4.4.3 of MIL-E-0022200E. One sample shall be subjected to all the tests specified in table VII (i.e., x-ray, mechanical and complete chemical analysis). The samples representing each of the remaining wet mixes of covering mixture in each lot shall be subjected to equivalency tests as specified in table VIII.

Table VII - Quality conformance inspection.

Electrode type	Required tests	Procedures
All	Test No. 3 and 8a 1/2/	Table X
All	Table XII	Table XII

1/ Types MIL-90XX-B3 and MIL-90XX-B3L electrodes shall be subjected to one rebaking at 800°F. ± 25°F. for a period of 1/2 hour to 1 hour prior to testing. It should be noted that this rebaking is in addition to the initial manufacturing baking of the electrode covering. This rebaking shall be accomplished except where specifically not required by the procurement document (see 6.2).

2/ Impact properties shall be determined only when specifically required by the procurement document.

Table VIII - Wet mix equivalency tests.

Electrode type	Required tests	Procedures
All All	Test 8a Chemical analysis 1/2/3/	Table X

1/ Required chemical analysis of the deposited weld metal is for selected elements only as specified in table IX.

2/ Equivalency chemical analysis tests are not required when the lot is as defined in 4.4.3.1(c) of MIL-E-0022200E.

3/ Paragraph 4.6.8.1 of MIL-E-0022200E may be applied.

Table IX - Wet mix equivalency weld deposit chemical analysis.

Electrode type	Required elements
All	Manganese Silicon Chromium Molybdenum

4.4.4 Unsatisfactory test results. If the results of the first tests are determined to be unsatisfactory, two retests shall be permitted. The results of both retests shall be satisfactory for quality conformance inspection.

4.5 Test procedures.

4.5.1 Weld tests. Table X summarizes the weld tests required under this specification for qualification testing of electrodes and for the inspection specified in tables VI, VII, VIII and IX.

4.5.1.1 Preheat, interpass, and postheat treatment. The preheat, interpass, and postheat treatment required for the weld tests are summarized in table XI.

4.5.1.2 Weld joint base plate. The weld joint base plate shall be medium steel conforming to MIL-S-22698 or ASTM A131, with the fusion faces of the groove and backing strip clad with a layer of weld metal. The cladding shall be deposited with the type of electrodes being tested. The cladding thickness shall be 3/16 inch minimum. The clad surfaces shall be machined or ground reasonably smooth before assembly of the test plate. When available, base plates having the same chemical composition as the deposited weld metal of the electrode being tested can be substituted for the clad plates.

4.5.2 Other tests. Table XII summarizes tests, in addition to those specified in table X required for qualification testing of the electrodes for the QPL and for quality conformance inspection, as applicable.

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Table X - Summary of required weld tests.[1/]

Weld test No.	Electrode type	Diameter sizes						Weld joint design procedure	Welding position[2/]	Tests			Chemical	Requirements
		3/32	1/8	5/32	3/16	7/32	1/4			X-ray[3/]	Tensile[4/]	Impacts[5/]		
3	All	X	X	X	X	X	X	Fig. 1 and notes	F	------	X	X	Chemical analysis[6/]	3.7.1 of MIL-E-0022200E and tables III and IV herein
8a	All	X ----	X ---	X ----	---- X	---- X	--- X	Fig. 2 and notes	V F	X X	----- -----	------- -------	---------- ----------	3.7.1 of MIL-E-0022200E

1/ For qualification testing of the electrodes for the QPL, all electrode types shall be tested using direct current (d.c.) reverse polarity (r.p.). In addition, MIL-XX16 and MIL-XX18 electrode types shall be tested using alternating current (a.c.). For comparison inspection, when required, and quality conformance inspection, all electrode types shall be tested using d.c., r.p.

2/ V = vertical, F = flat.

3/ When both tests (3 and 8a) are applicable radiography of test No. 3 will satisfy the requirements for test No. 8a for class 2 electrodes. If test No. 3 is radiographed, test No. 8a may be eliminated for class 2 electrodes.

4/ Includes ultimate, yield, elongation, reduction in area.

5/ As required by table III.

6/ May be determined from milling sample taken from ruptured ends of tensile specimens or drillings. In event of variances with composition requirements or conflicting results by testing activities, analysis shall be verified by procedures specified in 4.6.8 of MIL-E-0022200E.

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Table XI - Preheat, interpass, and postheat treatment.

Type	Temperature °F.		
	Minimum preheat and interpass thickness (inches)		Postheat 1/2/ ± 25°F.
	3/4 and under	Over 3/4	
MIL-80XX-B2 MIL-80XX-B2L	300	400	1300
MIL-90XX-B3 MIL-90XX-B3L	300	400	1350
MIL-502-XX MIL-502-XX-L	400	500	1375
MIL-505-XX MIL-410-XX	400 400	500 500	1375 1375

1/ For types MIL-80XX-B2L, MIL-90XX-B3L and MIL-502-XX-L, required only for qualification testing of these electrodes for the QPL (see notes to table III.)

2/ For postheat treatment, the holding time at temperature shall be 1 hour per inch of thickness. For thicknesses less than 1 inch the holding time shall be proportional to the thickness but not less than 30 minutes. The maximum rate of raising or lowering the temperature shall be 200°F. per hour.

Table XII - Summary of other tests.

Tests	Procedures	Requirements
	MIL-E-0022200E	
Visual and dimensional	4.6.2.3	3.10 (MIL-E-0022200E) and 3.6 herein
Concentricity	4.6.2	3.5.2 (MIL-E-0022200E)
Dielectric strength 1/	4.6.3	3.5.4 (MIL-E-0022200E)
Covering, flaking and cracking 1/	4.6.4	3.5.5 (MIL-E-0022200E)
Covering moisture	4.6.9	3.2.2.2 herein
Total iron content in covering 1/	4.6.10	3.2.2.1 herein

1/ Required for qualification (QPL) inspection only.

4.6 Certificate of test. A certificate of quality conformance of the weld tests specified in 4.4.3 (i.e., x-ray, mechanical, and chemical analysis), containing actual quantitative test and inspection results including the range of amperage employed in making the required welds for quality conformance inspection for each accepted lot, heat, and wet mix, as applicable, included in a particular shipment shall be furnished to the consignee with the shipment. The certification shall include test results from the quality conformance test and the wet mix equivalency test as applicable for the shipment. The minimum quality conformance test result data and wet mix equivalency test result data required shall be as shown on figures 3 and 4 and shall be submitted on the forms shown.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Packaging.

5.1.1 Level A. All electrodes shall be packaged in class 1a unit containers in accordance with MIL-W-10430, except that when the large containers are specified by the contract or order, 12, 14, and 18-inch electrodes may be supplied in class 1b unit containers.

5.2 Packing.

5.2.1 Levels A and B. Electrodes shall be packed level A or B as specified (see 6.2) in accordance with MIL-W-10430, except that for levels A and B packing, palletization for other than cylindrical cans shall be in accordance with MIL-STD-147.

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5.2.2 Level C. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations, as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.3 Marking.

5.3.1 Interior packages. Shipment marking information shall be provided on interior packages in accordance with the manufacturer's commercial practice. In addition to the lot identification required by 4.4.6 of MIL-E-0022200E, the information shall include MIL type, size, specification number, manufacturer's or distributor's name, date of manufacture (month/year), and manufacturer's or distributor's brand or type designation. All packages, or the smallest integral unit within a shipping container, shall carry the following caution label, as a minimum, prominently displayed in legible type on the package:

CAUTION
Welding may produce fumes and gases hazardous to health.
Avoid breathing these fumes and gases.
Use adequate ventilation.
See ANSI Standard Z49.1 "Safety in Welding and Cutting" published by the American Welding Society.

Significant toxic constituents when present in the electrode in greater than trace amounts shall be identified on the caution label so that normal ventilation can be increased accordingly. These constituents include but are not limited to those itemized in section 8.5 through 8.11 of ANSI Z49.1.

5.3.2 Exterior shipping containers and palletized unit loads. Shipment marking information shall be provided on exterior shipping containers and palletized unit loads in accordance with the manufacturer's commercial practice. In addition to the information required for interior packages as specified in 5.3.1, the information shall include shipping destination, Federal stock number, customer's order number, customer's item number, and customer's name.

6. NOTES

6.1 Intended use. This specification is intended to cover several "low hydrogen" types of arc welding electrodes of various alloys for elevated temperature service and for corrosion resistance. The designation "low hydrogen" applies to electrodes having coverings which are substantially free from water and other compounds such as cellulose which will liberate hydrogen in the arc. Intended use of each type is as specified in 6.1.1 through 6.1.6.

6.1.1 Type MIL-80XX-B2 is intended for welding 1-1/4 percent chromium, 1/2 percent molybdenum steel in high temperature service up to 950°F.

6.1.2 Type MIL-90XX-B3 is intended for the welding of 2-1/4 percent chromium, 1 percent molybdenum steel in high temperature service up to 1050°F. involving moderate conditions of corrosion or oxidation.

6.1.3 Type MIL-410-XX is intended for the welding of 12 percent chromium steel in service up to 1300°F. where very good resistance to corrosion and oxidation is required. Where strength is a factor, it is seldom used above 1200°F.

6.1.4 Type MIL-502-XX is intended for the welding of 5 percent chromium, 1/2 percent molybdenum steel in high temperature service up to 1200°F. involving severe corrosive or oxidizing conditions.

6.1.5 Type MIL-505-XX is intended for the welding of 9 percent chromium, 1-1/4 percent molybdenum steel in high temperature service up to 1300°F. involving severe corrosive and oxidizing conditions. This alloy has good oxidation resistance up to 1500°F.

6.1.6 Low-carbon type. The L type electrode is intended for use in welding materials of the same composition that the respective higher carbon type is used. L types may be used where preheat and interpass temperature control is impractical, where the base material is 1 inch thick or less. The L types also may be used on heavier than 1 inch base material with preheat and interpass temperature control 100°F. to 200°F. below that required for the respective higher carbon types. The L types may also be used on base material less than 1 inch in thickness when post heat is impractical.

6.2 Ordering data. Procurement documents should specify the following:

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(a) Title, number and date of this specification.
(b) Type, class and size required (see 1.2).
(c) For 3/32, 1/8, and 5/32 size electrodes, the length required (see table V).
(d) If rebaking is not required for quality conformance testing (see table VII, note 1).
(e) Level of packing required (see 5.2.1).

6.2.1 Electrodes should be ordered by the pound.

6.3 Inspection after delivery. Post delivery inspection of electrodes to determine conformity to this specification and for acceptance thereof is the responsibility of the consignee.

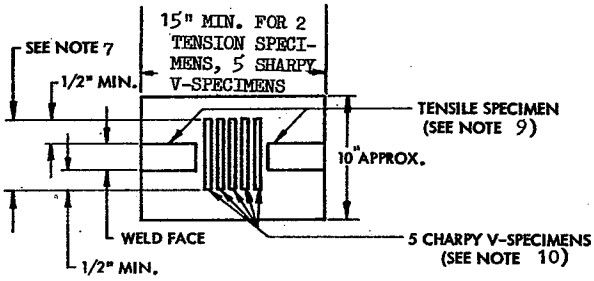
6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when materials and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Classification comparison. The classification comparison of this specification with classes specified in MIL-E-16589B is as follows:

MIL-E-16589B	This specification (MIL-E-0022200/8(SHIPS)
MIL-52-15 (1-1/4 CR-1/2 Mo) MIL-52-16 (1-1/4 CR-1/2 Mo)	MIL-8015-B2 MIL-8016-B2 MIL-8018-B2
MIL-52-LC-15 (1-1/4 CR-1/2 Mo) MIL-52-LC-16 (1-1/4 CR-1/2 Mo)	MIL-8015-B2L MIL-8016-B2L
MIL-82-15 (2 CR-1/2 Mo) MIL-82-16 (2 CR-1/2 Mo)	Deleted Deleted
MIL-82-LC-15 (2 CR-1/2 Mo) MIL-82-LC-16 (2 CR-1/2 Mo)	Deleted Deleted
MIL-94-15 (2-1/4 CR-1 Mo) MIL-94-16 (2-1/4 CR-1 Mo)	MIL-9015-B3 MIL-9016-B3 MIL-9018-B3
MIL-94-LC-15 (2-1/4 CR-1 Mo) MIL-94-LC-16 (2-1/4 CR-1 Mo)	MIL-9015-B3L MIL-9016-B3L
MIL-202-15 (5 CR-1/2 Mo) MIL-202-16 (5 CR-1/2 Mo)	MIL-502-15 MIL-502-16
MIL-202-LC-15 (5 CR-1/2 Mo) MIL-202-LC-16 (5 CR-1/2 Mo)	MIL-502-15-L MIL-502-16-L
MIL-365-15 (9 CR-1-1/4 Mo) MIL-365-16 (9 CR-1-1/4 Mo)	MIL-505-15 MIL-505-16
MIL-480-15 (12 CR) MIL-480-16 (12 CR)	MIL-410-15 MIL-410-16

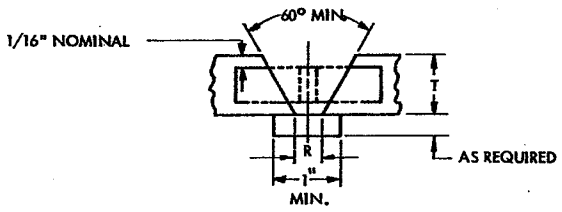
Preparing activity:
Navy - SH
(Project 3439-N223)

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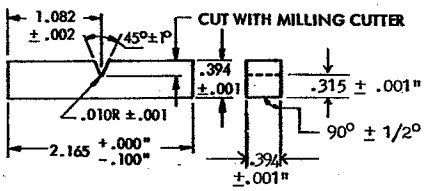
LAYOUT OF TEST PLATE.

NOTE: LENGTH MAY BE DECREASED ACCORDINGLY WHERE IMPACT SPECIMENS ARE NOT REQUIRED.



GROOVE PREPARATION OF TEST PLATE

SHOWING ORIENTATION OF CHARPY-V TEST SPECIMEN



CUT NOTCH WITH MILLING CUTTER. MAKE FOUR CUTS PER NOTCH IN FOLLOWING ORDER: a) .040; b) .025; c) .010; d) .004. NO IRREGULARITIES IN CONTOUR OF APEX OF NOTCH. NO. 8 POLISH MICRO-INCH FINISH.

Figure 1 - Welded joint for all-weld-metal tension test and impact test (test No. 3).

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Notes to figure 1:

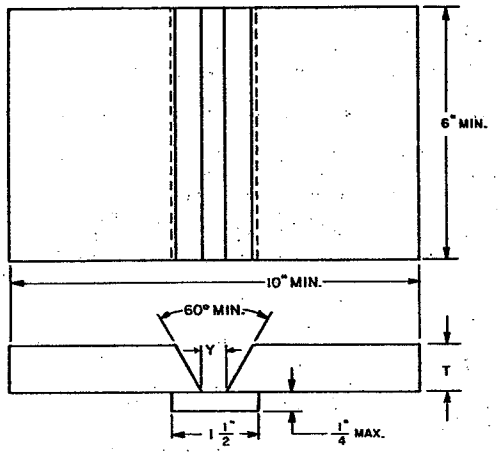
1. Base metal shall be as specified in 4.5.1.2.
2. Types MIL-90XX-B3 and MIL-90XX-B3L shall be qualification tested and quality conformance tested after rebaking in accordance with the procedure outlined in note 1 of table VII unless specifically exempted by the procurement document.
3. For qualification of the electrodes for the QPL, impact tests are required for types MIL-8018-B2 and MIL-9018-B3.
4. Welds shall be deposited in the flat position using d.c., r.p. and a.c. as required by note 5 to table X. The actual amperage values used shall be within the manufacturer's recommended current range and shall be recorded and reported. The root weld layer may be deposited with 1/8-inch diameter electrodes when testing 3/16-inch and larger diameter electrodes, provided both electrodes have the same manufacturer's designation. Welds shall be deposited in approximately 1/8-inch thick layers using a 2-1/2 core wire diameter maximum weave buildup sequence. The final layer shall include a reinforcement of standard proportions (1/32-inch to 1/8-inch). Each layer shall be started at the finishing end of the preceding layer. The operator error, pass and layer sequence shall be recorded and reported, as specified in table XIII.
5. After the joint has been tacked, the assembly shall be preheated as required by table XI. After completion of each layer the assembly shall be allowed to cool in still air at room temperature and the subsequent layer shall be started where the temperature of the plate has dropped to the interpass temperature of table XI. If it is necessary to interrupt this procedure, the assembly shall be slow cooled and preheated again.
6. Radiographic examination of test No. 3 is not required, but may be used in lieu of radiography of test No. 8a for class 2 electrodes when test No. 3 is radiographed. When used, the weld reinforcement and the backing shall be removed flush with the base plate surfaces. The assembly shall be radiographed in accordance with MIL-STD-271. Radiographs shall meet acceptance standards of NAVSHIPS 0900-003-9000 as specified in 3.7.1 of MIL-E-0022200E.
7. The test weldment shall then be cut into tensile and impact coupons as required by the notes to table III. No base metal shall be removed within 1/2-inch of the edge of the face of the weld by flame cutting. Only sawing or machining shall be used within 1/2-inch of the weld.
8. Coupons shall be stress-relieved when required in an electric furnace and in accordance with table XI and notes thereto, except when the whole test weldment was previously stress-relieved (see note 6 above).
9. The as-welded and stress-relieved tensile coupons shall be machined into one all-weld-metal tensile specimen for each test required by table III and notes thereto. The specimens shall be taken from the locations detailed on figure 1. The stress-relieved tensile coupon shall be etched, when necessary, to establish the location of the weld. The 0.505-inch diameter specimens shall be machined from 3/4-inch thick coupons and 0.357-inch diameter specimens from 1/2-inch thick coupons. See MIL-STD-418 or ASTM A370 for details. The specimens shall be tested for conformance with table III.
10. The stress-relieved impact coupon shall be etched, when necessary, to establish the location of the weld. Five Charpy V-notch specimens shall be machined in accordance with the details shown on figure 1.
11. Impact properties for the five specimens shall be obtained at test temperatures specified in table III herein, plus or minus 3°F. The extreme lowest impact value thus obtained together with the extreme highest value may be disregarded for this test. Two of the three remaining values shall be greater than the specified 20 foot/pound energy level; one of the three may be lower but shall be not less than 15 foot/pounds; the computed average value of these three shall be greater than the specified 20 foot/pound level (see table III herein).

Table XIII - Groove weld preparation.

Electrode size	Minimum thickness (T)	Minimum root opening (R)	Number of layers	
			Min.	Max.
(Inch)	(Inch)	(Inch)		
3/32	1/2	1/4	1/	1/
1/8	1/2	1/4	1/	1/
5/32	3/4	1/2	7	9
3/16	3/4	1/2	7	9
7/32	3/4	1/2	7	8
1/4	3/4	1/2	9	11

1/ Number of layers shall be reported.

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Electrode size	3/32 (Inch)	1/8 (Inch)	5/32 (Inch)	3/16-5/16 (Inch)
Root opening (Y)[1/]	1/4	5/16	3/8	1/2
Thickness (T)[2/]	1/4	3/8	1/2	1/2
Weld bead Width (max.)	3/8	1/2	5/8	3/4

[1/] Tolerance ± 1/16-inch.
[2/] Minimum.

Figure 2 - Groove weld usability test (test No. 8a).

Notes to figure 2:

1. Welding shall be accomplished with the test plate in the vertical or flat position as specified in table X. Direct current reverse polarity and alternating current shall be used in accordance with note 1 to table X. The actual amperage values used shall be within the manufacturer's recommended current range and shall be recorded and reported.
2. Base metal shall be in accordance with 4.5.1.2 herein.
3. The edge preparation shall be flame cut and ground to remove slag and surface irregularities.
4. Welds shall be deposited in approximately 1/8-inch thick layers. Weld bead widths shall be as shown above. Each bead shall contain a start in the area to be evaluated.
5. The reinforcement shall be 3/16-inch maximum.
6. Preheat and interpass temperature shall be as specified in table XI.
7. After completion of the weld, reinforcement and backing strips shall be removed flush with the base plate on both surfaces and the assembly shall be x-rayed in accordance with MIL-STD-271.
8. The radiograph shall meet the x-ray standards of NAVSHIPS 0900-003-9000 as specified in 3.7.1 of MIL-E-0022200E.

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CERTIFICATION OF QUALITY CONFORMANCE TESTS

Manufacturer or Distributor_____ Customer's Name_____

Address_____

Date_____ Customer's Order No._____

Specification MIL-_____

Type MIL-_____ Core Wire Heat No._____

Diameter & Length_____ Lot Identification MIL-E-0022200E Para._____

Inspection Level_____

Lot No._____ Wet mix No._____

Chemical Analysis (Complete)

Carbon_____

Manganese_____

Phosphorus_____

Sulfur_____

Silicon_____

Chromium_____

Molybdenum_____

Mechanical Test	As-Welded	Stress-Relieved
Yield Strength (0.2% offset method)		
Tensile Strength		
Elongation (%)		
Reduction of Area (%)		

Chemistry was taken from: ☐ Chem Pad ☐ Groove Weld

X-Ray Results_____ Groove Weld Test

Concentricity (%)_____

Test No.	3	8a	Chem Pad
Amperage			

Covering Moisture_____

Covering Iron Content_____ Operator Error (Layer Nos.)_____

Grinding During 8a Test Plate Preparation 3/

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____

NOTES:

1. Items not applicable to the type electrode involved or exempted by the procurement document shall be marked N/A.
2. This form is applicable to the quality conformance tests required for each lot. Figure 4 forms shall be employed to report and certify the wet mix equivalency tests where applicable for each wet mix of covering mixture in the lot. One copy of the figure 4 form shall be employed for certifying equivalency of each wet mix in a lot where applicable. One copy of the quality conformance certificate (figure 3) for the lot shall be forwarded with copies of the mix equivalency certificates for each additional mix from that lot that is included in the shipment.
3. See paragraphs 3.5.7 and 3.9 of MIL-E-0022200E.

Figure 3 - Certification of quality conformance tests.

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CERTIFICATION OF WET MIX EQUIVALENCY TESTS

Manufacturer or Distributor_____ Customer's Name_____

Address_____

Date_____ Customer's Order No._____

Specification MIL-_____

Type MIL-_____

Diameter Size_____ Core Wire Heat No._____

Lot No._____ Wet mix No._____

Sampling
Chemical Analysis

Chemistry was taken from:

Manganese_____ Chem Pad Groove Weld

Silicon_____ Test No._____

Chromium_____ Chem Pad 8a

Amperage _____ _____ _____

Molybdenum_____

X-Ray Results_____

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____

NOTES:

1. Items not applicable to the type electrode involved or exempted by the procurement document shall be marked N/A.
2. One copy of the figure 4 form shall be employed for certifying equivalency of each wet mix in a lot where applicable. One copy of the quality conformance certificate (figure 3) for the lot shall be forwarded with copies of the mix equivalency certificates for each additional mix from that lot that is included in the shipment.

Figure 4 - Certification of wet mix equivalency tests.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

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Hyattsville, Maryland 20782

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